



# 人工智能時代的政府翻譯：最新發展趨勢與實踐策略

蕭世昌（GILT（全球化、國際化、本地化和翻譯）研究中心）

2026年6月5日

# 用 AI 翻譯時遇過以下情形嗎？

- 譯文生硬死板
- 譯名前後不一致
- 虛構姓名、地名、政策名稱
- 風格和格式不對，要大幅改寫
- 長文件漏譯整句整段
- 聲稱「已查證官方資料」（其實沒有）
- 同一段原文，每次譯出來都不一樣
- AI 不遵循指示或者突然失憶，甚至把之前改好的地方又改錯了

# 大綱

## AI 翻譯2026年最新發展趨勢

## AI 如何協助翻譯工作：政府翻譯的實踐策略

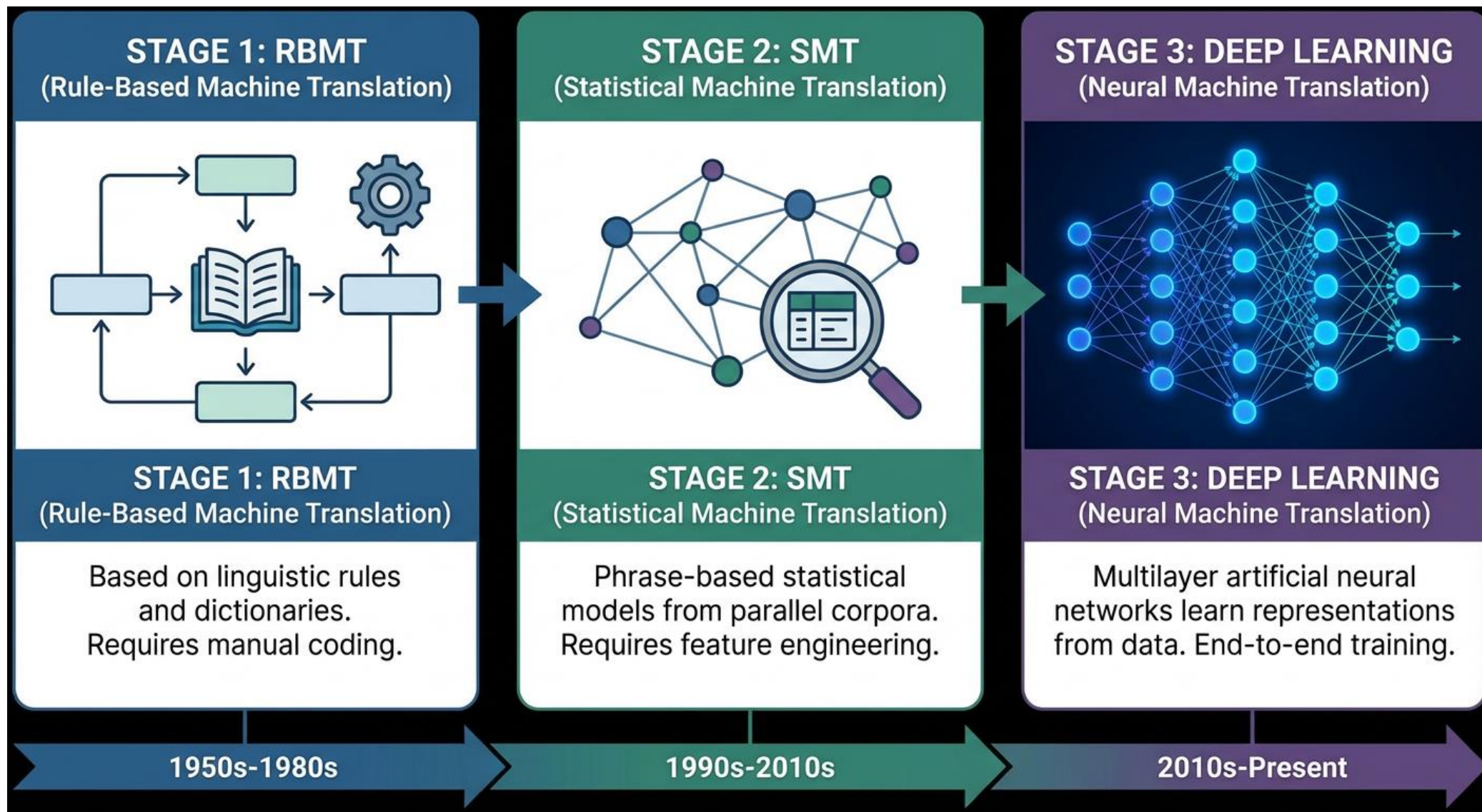
- 了解模型種類
- 掌握提示詞工程（即怎樣提問）和互動編輯技巧
- 善用相關資源
- 持續關注AI發展和發掘新技巧



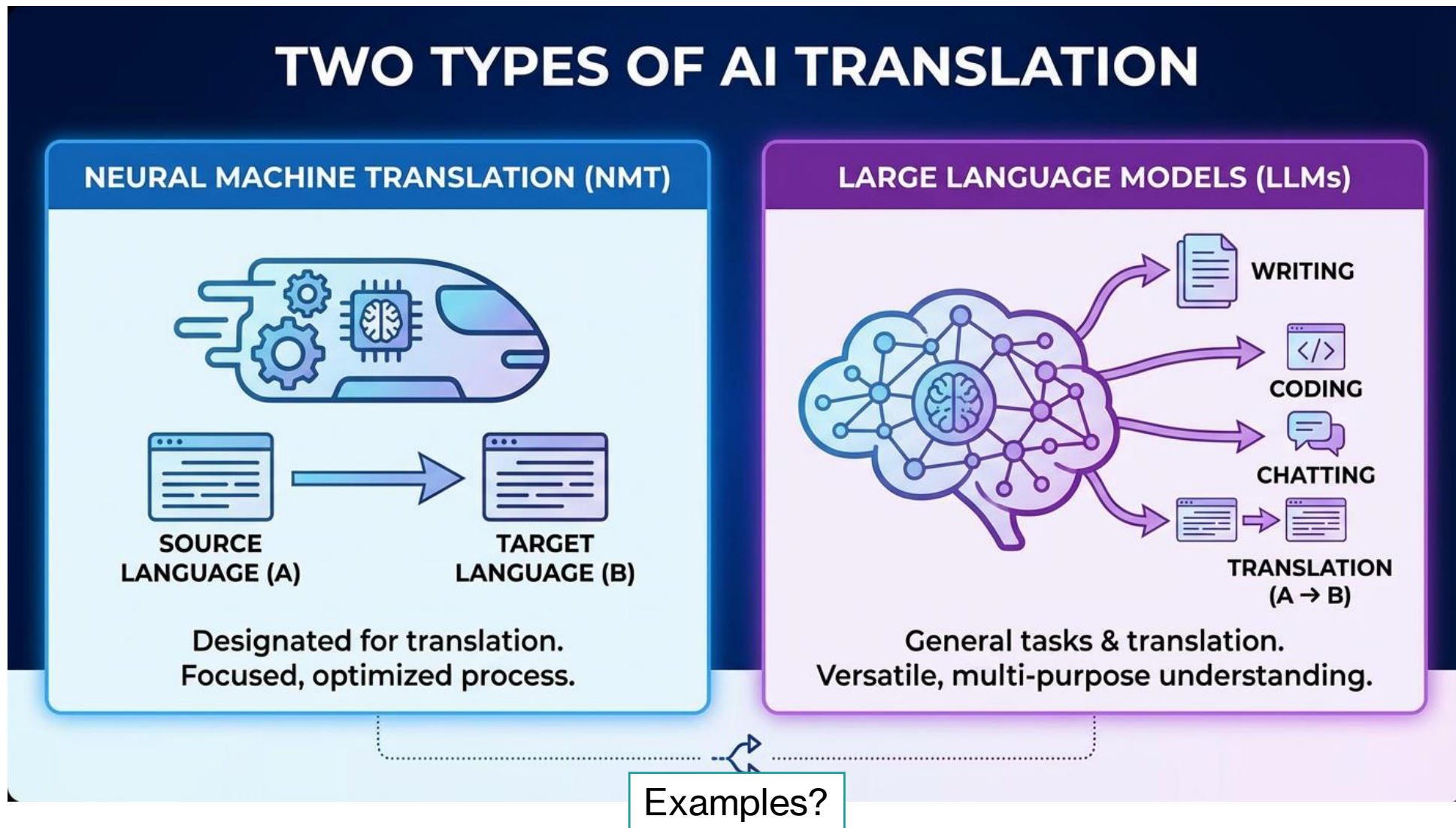
# I. AI 翻譯2026年 最新發展趨勢



# 自動翻譯的三大發展階段：規則、統計、深度學習



# 人工智能翻譯：神經機器翻譯和大語言模型



**兩者最重要區別？**



# 大模型為何看懂提示詞：預訓練、微調、推理

## Key Steps in Modern AI Development & Student Analogy

AI DEVELOPMENT

### 1. Pre-training



The model learns from vast amounts of text (books, websites, legal documents, code)

### 2. Fine-tuning

SFT (Supervised Fine-Tuning)



Teaching the model specific tasks with labeled examples

RLHF (Reinforcement Learning from Human Feedback)



Humans rank outputs; the model learns human preferences

### 3. Inference



The moment you ask a question and the model generates an answer based on learned probabilities

STUDENT ANALOGY

### STAGE 1: STUDYING



Reading widely & gathering knowledge.

### STAGE 2: PRACTICING

SFT Analogy:



Seeing how others answer questions.

RLHF Analogy:



Answering / solving questions and teacher ranking.

### STAGE 3: TAKING EXAM



Demonstrating understanding under pressure.



# 神經機器翻譯與大語言模型：發展歷程

## Development of NMT and LLMs: A Converging Path

### EARLY DEEP NEURAL NETWORKS FOR TRANSLATION



2013/2014

Pioneering work in BiLSTM & CNN for NMT (e.g., Sutskever et al., Cho et al.) - Moving beyond SMT.



BiLSTM CNN



2017

### THE TRANSFORMER ARCHITECTURE

Paper: "Attention Is All You Need" - Revolutionized NMT, enabled massive parallelization, and opened the era of LLMs.



#### GPT-1

First GPT, introducing self-supervised learning on large data, pre-training then fine-tuning.



#### GPT-2

Scaled-up parameter count (1.5B), demonstrated zero-shot capabilities, raised concerns about misuse.



#### 2020 GPT-3

Massive scale (175B parameters), impressive few-shot and zero-shot performance, powerful text generation.



#### 2022 GPT-3.5 (ChatGPT)

Fine-tuned for dialogue with RLHF, bringing AI to the public mainstream.



#### 2023 GPT-4

Multimodal capabilities (image/text input), enhanced reasoning, performance on academic and professional benchmarks.



#### GPT-4o & o1 (Reasoning Models)

Faster, more efficient (Omni), and enhanced reasoning capabilities (o1) for complex problem-solving.



#### GPT-5 & GPT-5.1

Expected to bring even greater scale, improved reasoning, multi-agent capabilities, and further integration.

2017  
Transformer for NMT

2018-2022  
Advanced NMT architectures & Multilingual models (e.g., mBART, T5)

2023+  
Convergence with LLMs for translation tasks

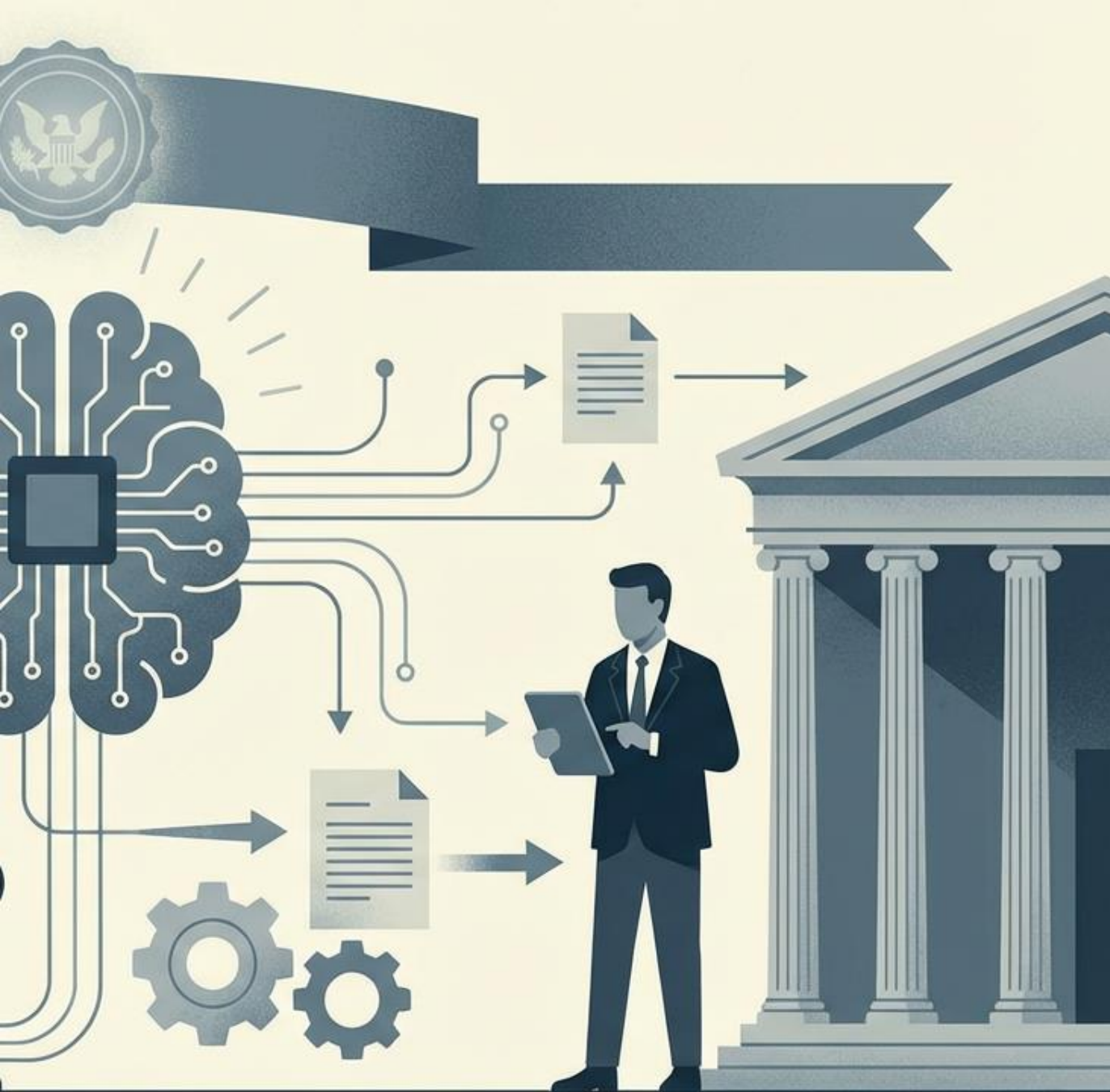
# 最新模型選錄 ( 2026年6月3日 )

| Model Name           | Company / Developer | URL                                                                     |
|----------------------|---------------------|-------------------------------------------------------------------------|
| GPT-5.5              | OpenAI              | <a href="https://openai.com/">https://openai.com/</a>                   |
| Gemini-3.5-Flash     | Google              | <a href="https://gemini.google.com/">https://gemini.google.com/</a>     |
| Claude Opus 4.8      | Anthropic           | <a href="https://www.anthropic.com/">https://www.anthropic.com/</a>     |
| Grok 4.3             | xAI                 | <a href="https://x.ai/">https://x.ai/</a>                               |
| Perplexity           | Perplexity AI       | <a href="https://www.perplexity.ai/">https://www.perplexity.ai/</a>     |
| DeepSeek V4          | DeepSeek            | <a href="https://www.deepseek.com/">https://www.deepseek.com/</a>       |
| Kimi 2.6             | Moonshot AI         | <a href="https://kimi.moonshot.cn/">https://kimi.moonshot.cn/</a>       |
| Qwen 3.7             | Alibaba Cloud       | <a href="https://www.qianwen.com/">https://www.qianwen.com/</a>         |
| Llama 4              | Meta                | <a href="https://www.llama.com/">https://www.llama.com/</a>             |
| Doubao-Seed-2.0 (豆包) | ByteDance           | <a href="https://www.doubao.com/">https://www.doubao.com/</a>           |
| Metaso (秘塔)          | Metaso              | <a href="https://metaso.cn/">https://metaso.cn/</a>                     |
| GLM-5.1 (智谱清言)       | Zhipu AI            | <a href="https://chatglm.cn/">https://chatglm.cn/</a>                   |
| Ernie 5.1 (文心一言)     | Baidu               | <a href="https://yiyan.baidu.com/">https://yiyan.baidu.com/</a>         |
| SenseNova-V6.5 (商量)  | SenseTime           | <a href="https://chat.sensetime.com/">https://chat.sensetime.com/</a>   |
| Yuanbao (騰訊元寶)       | Tencent             | <a href="https://yuanbao.tencent.com/">https://yuanbao.tencent.com/</a> |
| Skywork (天工 AI)      | Skywork AI          | <a href="https://skywork.ai/">https://skywork.ai/</a>                   |

# 一站式AI模型平台

- Poe (<https://poe.com/>)





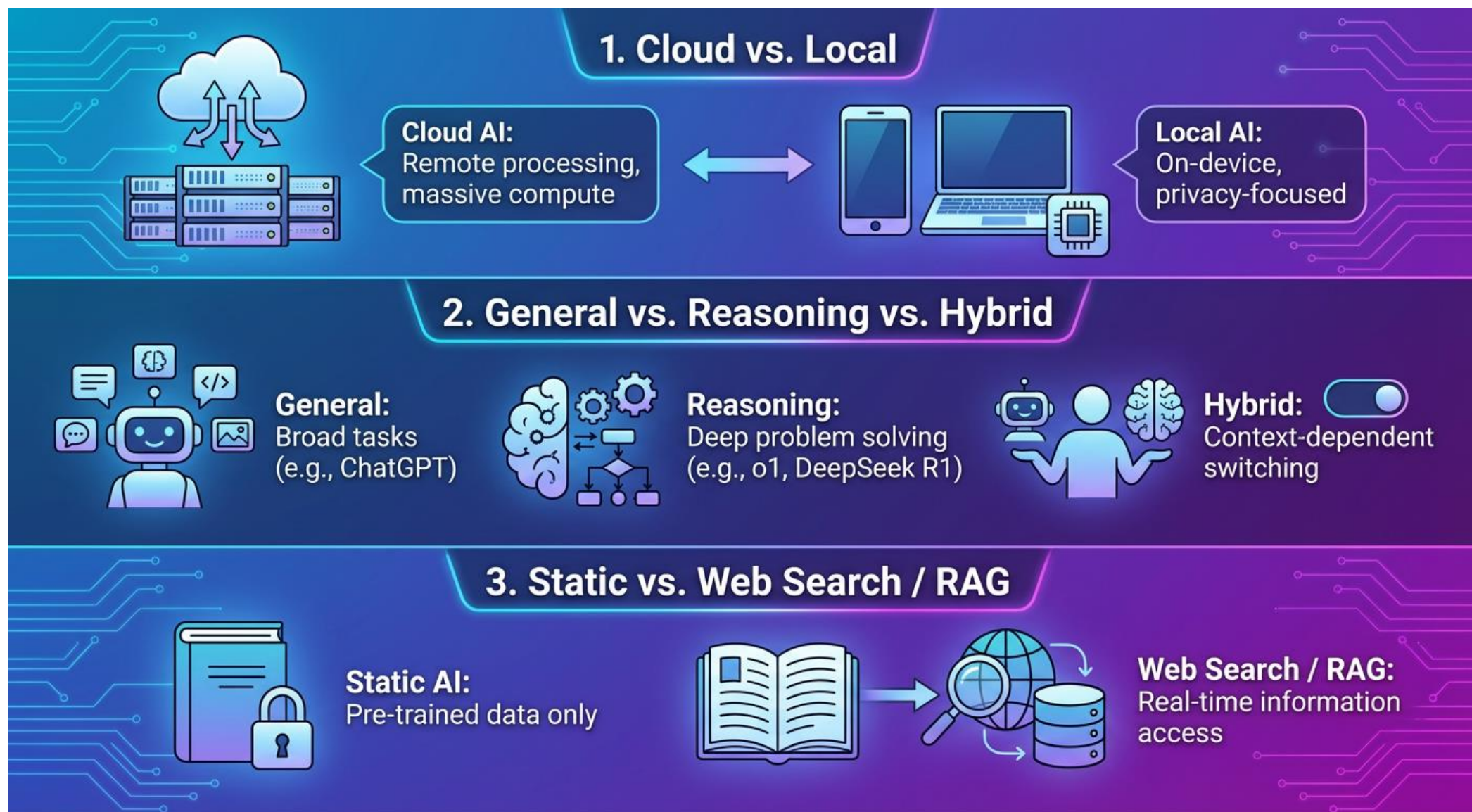
## II. AI 如何協助翻譯工作： 政府翻譯的實踐策略



# 1. 了解模型種類



# 不同種類的人工智能模型





# 其他重點

## Other Key LLM Selection Criteria



### Model Size (Parameters)

- **Large Models (100B+):** Superior nuance, reasoning, and knowledge depth. Best for complex legal & policy work.
- **Smaller Models (<70B):** Faster, efficient, can run on less powerful hardware. Excellent for summarization, initial drafting, or local fine-tuning.



### Reasoning & Accuracy

Evaluates the model's ability to handle complex logical, legal, and policy-based reasoning.

Test with multi-step instructions and nuanced text.



### Context Window Length

How much text the model "remembers" for maintaining consistency.

A large window (e.g., 200K+ tokens, ~300+ pages) is essential for long documents.



### Data Cutoff & Web Search

For texts on recent events, a recent cutoff date or live web search is crucial.

**Warning:** Always verify web-sourced information against official sources.

*Selection involves balancing these factors based on specific use cases and resource constraints.*



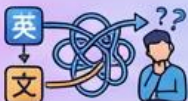
## 2. 掌握提示詞工程 和互動編輯技巧



# 人工智能翻譯的常見問題

## Common Issues in AI Translation: Understanding Flaws for Effective Post-Editing

Understanding these flaws is the first step to effective post-editing.

|  Issue                                                  |  Description                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>1. Incomplete Translation</b>                       | Dropping sentences or paragraphs, especially in long texts. Model may summarize instead of translating verbatim.                                                                    |
|  <b>2. Wrong Terms / Hallucination</b>                  | Inventing official names, dates, legal definitions, or statistics (e.g. wrong government official name?).                                                                           |
| <br>word-for-word <b>3. Monotonous / Homogenization</b> | <b>"AI Translationese"</b> . Overly literal adherence to source text structure. Lacking linguistic variety.                                                                         |
|  <b>4. Term/Style Inconsistency</b>                    | AI has output its own style – different from in-house style guides, previous documents, or text-type conventions. May translate the same term differently within the same document. |
|  <b>5. Unnaturalness (Eng → Chi)</b>                  | <b>Europeanized Chinese</b> (like long noun phrases affecting readability?). LLMs often 'think' in English logic, leading to unnatural Chinese phrasing.                            |
|  <b>6. Randomness / Non-Determinism</b>               | Every generation yields slightly different results due to temperature settings and token sampling.                                                                                  |



# 策略 1. 人手潤色費心機？工具輔助顯優勢

## USEFUL EDITING TOOLS



### DeepL Write

- › AI-powered writing assistant for clarity & style.



### QuillBot

- › Paraphrasing and summarizing tool.



### NeuralWriter

- › Multilingual content generation and rephrasing.



### Ahrefs Paraphrasing Tool

- › SEO-focused sentence rewriter.

# 例子

- 為配合國家的教育數字化策略，教育局持續於中小學大力推動數字教育，提升學生的數字素養與技能。行政長官今年在《施政報告》就加強推動中小學數字教育，提出多項新的政策措施，包括在優質教育基金預留**20**億元支援中小學數字教育、制定中小學數字教育藍圖、加強教師人工智能培訓，以及引入企業資源等。（[來源](#)）
- **AI 翻譯（GPT-5.1）**：To align with the country's strategy for digitalizing education, the Education Bureau has been continuously and vigorously promoting digital education in primary and secondary schools, in order to enhance students' digital literacy and skills. In this year's Policy Address, the Chief Executive proposed a number of new policy measures to further promote digital education in primary and secondary schools. These include reserving 2 billion dollars under the Quality Education Fund to support digital education in primary and secondary schools, formulating a digital education blueprint for primary and secondary schools, strengthening teachers' training in artificial intelligence, and introducing corporate resources, among others.

## 策略 2. 譯後編輯多煩惱？譯前工作要做好

### Proactive Translation Quality: Pre-Editing & Prompt Engineering

Don't wait for post-editing; improve quality before and during the LLM process.

#### STRATEGY 1: PRE-EDITING THE SOURCE TEXT



**1.1 Use Editing Tools Before Prompting:** Apply grammar, style, and clarity tools to the original text.



**1.2 Comprehensive Source Text Analysis & Preparation:**



**Analyze for Key Points:** Identify core message and target audience.



**Summarize or Visualize Content:** Create outlines or visual aids.



**Check for Existing Issues:** Verify facts and correct factual errors.



**Identify Key Terms:** Create a glossary for consistency.

CLEANER SOURCE

OPTIMIZED  
LLM INPUT

#### STRATEGY 2: ENHANCING YOUR TRANSLATION PROMPT



**2.1 Define Roles & Context:** Specify persona, audience, and purpose.



**2.2 Step-by-Step Thinking:** Ask the LLM to analyze the text or outline its plan before translating.



**2.3 Controlled Chunking:** For long texts, break into segments to prevent summarization and ensure detail.



**2.4 Context Reinforcement ("Don't Forget"):** Regularly remind the LLM of core instructions in multi-turn conversations.

CLEARER INSTRUCTIONS



**SUPERIOR LLM  
TRANSLATION  
OUTPUT**  
(Less Post-Editing  
Required)



# 譯前編輯

- 譯後編輯若效果欠理想，或許可從原文入手
- 英漢翻譯
  - 譯前編輯：英文改寫工具
  - 譯後編輯：中文改寫工具
- 漢英翻譯
  - 譯前編輯：中文改寫工具
  - 譯後編輯：英文改寫工具

# 演示

- **英漢翻譯**：The Cultural and Creative Industries Development Agency (CCIDA) under the Culture, Sports and Tourism Bureau (CSTB) has organised the “Hong Kong Cultural and Creative Mosaic” event at the Palace Museum Cultural and Creative Products Hong Kong Space inside the Palace Museum in Beijing. The event officially opened to the public today (November 20) to stage four successive events during the year ahead to showcase cultural and creative products with themes of Hong Kong’s intangible cultural heritage, art toys, fashion and accessories, and lifestyle products. ( [來源](#) )
- **漢英翻譯**：為配合國家的教育數字化策略，教育局持續於中小學大力推動數字教育，提升學生的數字素養與技能。行政長官今年在《施政報告》就加強推動中小學數字教育，提出多項新的政策措施，包括在優質教育基金預留**20**億元支援中小學數字教育、制定中小學數字教育藍圖、加強教師人工智能培訓，以及引入企業資源等。  
( [來源](#) )

# 翻譯提示詞工程

- 譯後編輯若效果欠理想，也可以從翻譯提示詞入手
- 方法一：定義角色，並且提供與當前翻譯任務有關的不同資料
- 方法二：引導模型一步一步思考才提供翻譯（思維鏈提示詞）
- 方法三：要求模型分批輸出結果
- 方法四：適當時重新輸入提示詞，以防模型忽然「失憶」



# 翻譯提示詞工程

- 例1: 原文檢查
- 例2: 角色與情境設定 ( 草擬副手 )
- 例3: 進階思維鏈 (CoT)
- 例4: 分批輸出與長文處理
- 例5: 防失憶與情境重置

# 策略 3. 譯文優劣怎認定？貨比三家要看清

**Compare Different Results:** Never settle for the first draft. Generate and compare multiple options.

## 1. Single LLM, Multiple Outputs



## 2. Multiple LLMs (Poe's /compare)



## 3. Diff Tools



## 4. Use the LLM as Judge



# 要求模型提供多個翻譯結果

- 方法一：利用提示詞
- 示例: 多版本生成
- 方法二：使用 `/repeat [Num]` 指令（適用於Poe）



# 使用不同模型進行翻譯

- 使用以下指令：`/compare @ModelName`（適用於 Poe）
- 特別技巧：結合內地和國外模型

# 使用文本對比工具

若干例子

- Microsoft Word
- Google Docs
- CountWordsFree – Compare Text Online  
(<https://countwordsfree.com/comparetexts>)
- Draftable (<https://www.draftable.com/compare>)

# 策略 4. 術語不一難確認？官方權威有保證

## Verify Terms & Official Information

Never trust the AI with legal terms or government names. Always verify.

### 1. Web Search Verification & URL Language Code Switching



#### Advanced Search Operators:

`</>` site:info.gov.hk  
`</>` site:legco.gov.hk  
`</>` site:elegislation.gov.hk  
`</>` site:hkexnews.hk

#### URL Language Code Switching

https://www. `<url>` /eng/...   
https://www. `<url>` /tc/... 

Switching between  
/eng/ and /tc/

Use official government or legal sites for verification.

VERIFICATION  
PROCESS



### 2. Glossary-Based Verification



#### (a) Provide glossary in prompt and ask LLM to check

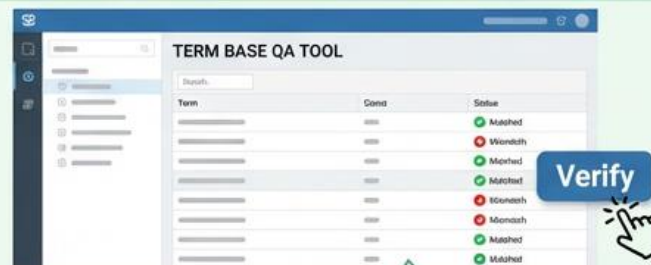


**Prompt:** Check against this glossary:  
[Term A: Definition A, Term B: Definition B]...

**LLM:** Term A is used correctly. Term B needs revision based on glossary.



#### (b) Use QA tool provided by term base systems



Leverage established terminology databases for consistency.

**Accuracy First:** Reliable sources and structured checks ensure trustworthy information.





# 譯前術語識別和提示詞處理

- 分析原文中的術語（ 輔助工具例子：  
<https://www.tomedes.com/tools/pre-translation> ）
- 將術語資訊添加到提示詞

# 術語相關提示詞示例

- 例1: 譯前術語提取
- 例2: 強制套用術語表
- 例3: 縮寫與機構名稱處理

# 譯後術語檢查

輔助工具例子：

- 例子一：<https://www.tomedes.com/tools/post-translation>
- 例子二：<https://www.matecat.com/>

# 譯後術語檢查

- 示例: 譯後術語掃描與除錯



# AI不能盡信

Translate the following text into English:

文化體育及旅遊局局長羅淑佩表示，昨日內地訪港旅客約有145,000人次，較上月周末平均每日數字高28.5%，令人振奮。她相信這是深圳「一簽多行」個人遊簽注恢復帶給香港的新機遇。

# 檢索官方資訊

- 例子一：Linguee (<http://www.linguee.com/english-chinese/>)
- 例子二：搜尋引擎的進階檢索功能

# 進階檢索的三大技巧

- 技巧一：雙引號
- 技巧二：指定來源
- 技巧三：修改URL

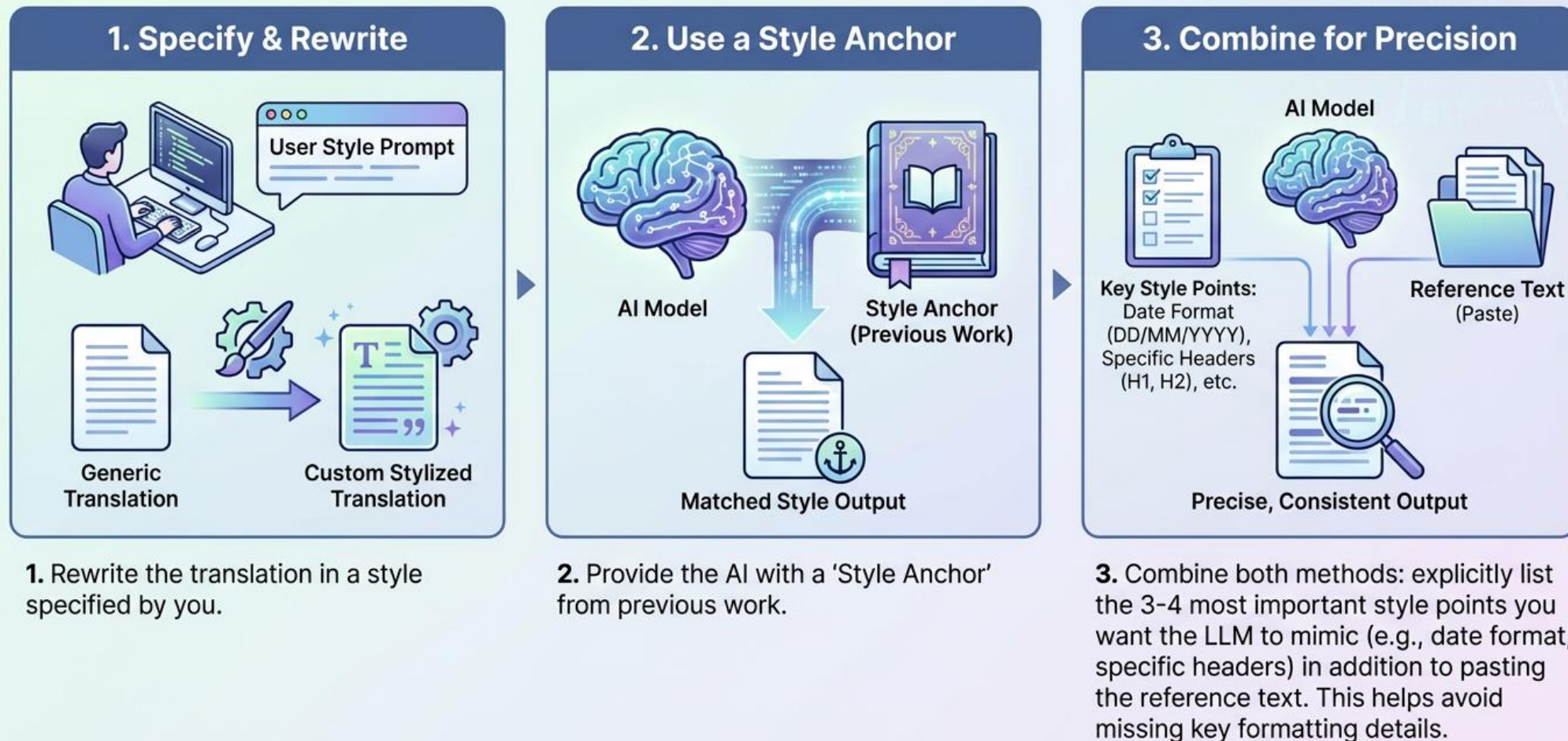
# 一些例子

- personalised vehicle registration marks
- The Administration would not preclude ...
- as soon as practicable
- consented to or voted in favour of the special resolution
- the Inside Information Provisions under Part XIVA of the SFO
- 上調人民幣貸款基準利率
- 製造及生產線「升」級支援先導計劃
- 和相關持份者保持緊密聯繫
- 如公司違反本條
- 除文義另有所指外



# 策略 5. 風格混亂惹投訴？譯文實例好法寶

## Achieving Style Consistency in AI Translation



# 在提示詞中指定翻譯風格

- 輔助工具：<https://www.tomedes.com/tools/style-guide-generator>
- 此工具讓用家提供原文，**AI**評估應採取的翻譯風格（之前4.1介紹的工具也提供類似功能）
- 附加建議 1：不妨提供一篇風格相近的譯文文字
- 附加建議 2：在提示詞添加譯例

# 在提示詞中指定翻譯風格

- 例1: 風格提取與分析
- 例2: 指定風格指南
- 例3: 範例引導翻譯 (Few-Shot)
- 例4: 週期性報告參考翻譯

# 策略 6. AI譯腔揮不去？英漢差異是精髓

## ENHANCED TRANSLATION CLARITY VIA CONTRASTIVE LINGUISTICS

Guiding AI away from "Translationese" using English-Chinese differences.

### SET A: ENGLISH → CHINESE (Avoiding "Europeanized Chinese")

Focus: Dynamic, Topic-Prominent, Clause-Rich



### SET B: CHINESE → ENGLISH (Avoiding "Chinglish")

Focus: Explicit Logic, Morphological Precision, Condensed Structure



**RESULT:** More Natural, Accurate, and Culturally Appropriate Translations.

# 六大要點

- 顯性與隱性邏輯：連接與銜接 (Logic & Cohesion)
- 主語轉換：物稱與人稱 (Inanimate vs. Animate Subjects)
- 級階轉換：句構階層與資訊密度 (Rank Shifting: Structure & Density)
- 動詞應用：動態轉換與連動結構 (Verb Application: Dynamics & Serial Verbs)
- 屈折變化的靈活處理 (Flexible Handling of Inflections)
- 搭配詞 (Collocations)



# 英漢互動編輯提示詞

- 例1：顯性與隱性邏輯
- 例2：主語轉換 (物轉人 / 主題)
- 例3：級階轉換 (解壓縮)
- 例4：動詞應用 (找回動詞)
- 例5：屈折變化 (去標記化)
- 例6：搭配詞 (歐化搭配)
- 例7：通用深度潤色 (去歐化)

# 漢英互動編輯提示詞

- 例1：顯性與隱性邏輯
- 例2：主語轉換 (人轉物)
- 例3：級階轉換 (壓縮)
- 例4：動詞應用 (連動詞 / 名詞化)
- 例5：屈折變化 (詞彙轉語法)
- 例6：搭配詞 (中式搭配)
- 例7：通用深度潤色 (語法精煉)

# 其他適用於譯後編輯或檢查的提示詞

- 例1：通用潤色與語氣校正
- 例2：針對性除錯：事實查核
- 例3：漏譯掃描
- 例4：標點與格式規範化
- 例5：譯後術語掃描與除錯
- 例6：譯後風格一致性檢查
- 例7：AI 擔任評審 (LLM as Judge)
- 例8：自我評分與品質估算（政府公文評分準則）

# 在對話中靈活運用各類提示詞

- 完整翻譯工作流程（英譯中示範）

# 其他探索方向

- 提示詞語言：中文還是英文？
- 語氣：客氣還是直接？
- 結構：段落式還是條列式？
- 要求**AI**用自己的話把提示詞內的要求重新講一遍





### 3. 善用相關資源

# 大語言模型排行榜

- Poe 排行榜：<https://poe.com/leaderboard>
- Vellum 排行榜：<https://www.vellum.ai/llm-leaderboard>
- LMArena 排行榜：<https://lmarena.ai/leaderboard>
- LLM：<https://llm-stats.com/>（提供context window資料）和  
<https://llm-stats.com/benchmarks>

# Markdown 標記語言轉換工具

- 各位看過以下的輸出格式嗎？

## ## Phase 1: Contextual Analysis

- **\*\*Direction:\*\*** English → Traditional Chinese (Hong Kong)
- **\*\*Text Type:\*\*** Press release / information note (cultural promotion activity)
- **\*\*Audience:\*\*** General public, media, cultural sector stakeholders, Mainland interlocutors
- **\*\*Tone:\*\*** Formal, informative, neutral, in line with HKSAR Government press releases

### Key style considerations:

- Use standard government names: 「文化體育及旅遊局」、「文化及創意產業發展署」.
- Mainland references follow usual practice: 「北京故宮博物院」、「香港故宮文化博物館」各有清晰區分，須留意原文語意。
- Concise but complete sentences, avoiding翻譯腔, matching gov.hk press-release style.

# Markdown 標記語言轉換工具

- 例子一：<https://markdowntohtml.com/>
- 例子二：<https://markdowntotext.com/>

# 其他AI文字處理工具

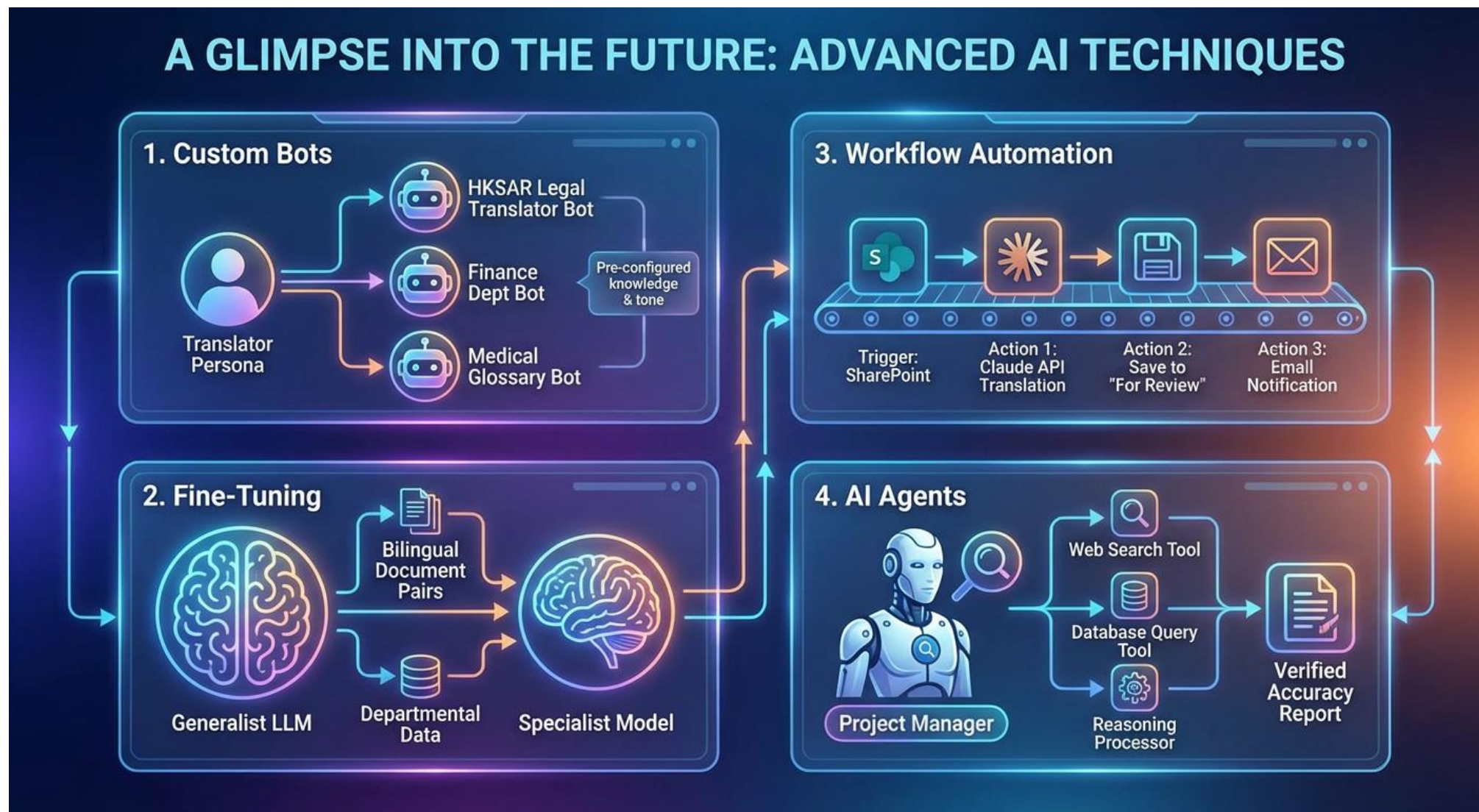
- 例子一：<https://machinetranslation.com/>
- 例子二：<https://www.tomedes.com/tools>
- 例子三：<https://ahrefs.com/writing-tools>





## 4. 持續關注AI發展和發掘新技巧

# 進階技巧和新興方向





# 了解更多

- Generative AI for Financial Translation: Opportunities and Challenges in Hong Kong (2026)  
([https://link.springer.com/chapter/10.1007/978-981-95-4374-8\\_6](https://link.springer.com/chapter/10.1007/978-981-95-4374-8_6))
- Revolutionising Translation with AI: Unravelling Neural Machine Translation and Generative Pre-trained Large Language Models (2024) ([https://link.springer.com/chapter/10.1007/978-981-97-2958-6\\_3](https://link.springer.com/chapter/10.1007/978-981-97-2958-6_3))
- Where Neural Machine Translation and Translation Memories Meet: Domain Adaptation for the Translation of HKSAR Government Press Releases (2024)  
(<https://www.taylorfrancis.com/chapters/edit/10.4324/9781003399261-7/neural-machine-translation-translation-memories-meet-siu-sai-cheong>)
- Deep Learning and Translation Technology (2023)  
(<https://www.taylorfrancis.com/chapters/edit/10.4324/9781003168348-50/deep-learning-translation-technology-sai-cheong-siu>)
- 部分論文備免費電子版，詳見此：<https://www.researchgate.net/profile/Sai-Cheong-Siu/research>

# 謝謝！

